Question			Answer/Indicative content	Marks	Guidance
1			Method A - E.g. Will not sample people who work then / people who do not walk down that street. Method B - E.g. This will only get answers from those who want to send in an answer. Method C - E.g. This will only get answers from those who use the council website. E.g. Those who use the internet more frequently are more likely to see the question.	B1(AO2.4) B1(AO2.4) B1(AO2.4) [3]	Enter text here.
			Total	3	
2		a	Allocate numbers 001 to 200 to the trees Choose 10 (3 digit) random numbers	B1(AO1.2) B1(AO2.4) [2]	e.g. use calculator to get 10 different random numbers
		b	Mean = 27.61kg SD = 4.04 kg (3sf)	B1(AO1.1) B1(AO1.1) [2]	BC BC
		c	Upper limit = 27.61 + 2 × 4.04 = 35.69 So the value of 38.1 is an outlier This value should be investigated to check if it is genuine. If so, it should not be removed from the data	M1(AO1.1) A1(AO1.1) B1(AO2.2b) [3]	For mean + 2 × sd OR UQ + 1.5 IQR = 28.3 + 1.5 × 3.2 = 33.1 OR e.g. If the value is not representative of the other 199 trees because e.g. this tree is a different type it should be ignored
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
3		a	Consistent use of midpoints in either calculation Mean £36.25 Sample sd 8.313… £36 250 and £8313	M1(AO1.1) A1(AO1.1a) A1(AO1.1) A1(AO1.1) [4]	soi BC BC FT their calculator values
		b	We are using grouped data not the original values	B1(AO2.4) [1]	
		c	Any valid reason which suggests that the sample is not necessarily representative	B1(AO3.2b) [1]	
		d	It would increase	B1(AO2.2a) [1]	
			Total	7	

Question			Answer/Indicative content	Marks	Guidance	
4		a	Population because these are all the countries of interest.	B1(AO1.1) [1]		
		b	Eg Consistent with the correlation for all countries oe And Eg You would expect countries with higher populations to tend to have higher numbers of both mobile phone subscribers and internet users. Or Eg people who use mobile phones will be more likely to use the internet	E1(AO2.4) E1(AO2.2b) [2]		
		c	Ukraine Has high mobile phone usage with lower internet provision. Suggests people are used to and/or like technology so potential customers for the internet.	B1(AO1.1) E1(AO3.2a) [2]		
		d	Number the companies from 000 to 599 or 001 to 600 Generate 3-digit random numbers (from a calculator or spreadsheet). Match number with number given to companies. [Discard 600 to 999.] Do not use any numbers twice.Stop when you have selected 20 different companies.	E1(AO1.2) E1(AO1.1) E1(AO2.4) [3]	Or from a table of random numbers. or discard 000 and 601 to 999.	**OR** Input list of companies into a spreadsheet **NB** Writing the names on paper. Putting these in a hat. Selecting 20. **E1 E1E0**. Not practical.
			Total	8		

Question			Answer/Indicative content	Marks	Guidance
5		a	(a) i Mean for world = 7 174 654 290 ÷ 239 = 30 million (approx.) Mean for sample is 9.36 million so much smaller	M1(AO 1.1) A1(AO 1.1) B1(AO 1.1) [3]	 Any degree of accuracy
		b	(a) ii There are a lot of small countries in the world. Therefore there is no reason to suppose the sample is not random,	B1(AO 2.4) E1(AO 2.2b) [2]	
		c	(b) 0.091 × 55 400 =[$] 5041.40	M1(AO 3.3) A1(AO 1.1) [2]	Condone missing units. Answer can be rounded to 5041 or 5040
		d	(c) i Identify physicians per 1000 as highest R^2 with positive correlation and positive correlation	B1(AO 2.2b) E1(AO 2.4) [2]	NB graph C e.g. Reject D as it shows negative correlation
		e	(c) ii There is hardly any correlation.	E1(AO 3.5b) [1]	
			Total	10	

Question			Answer/Indicative content	Marks	Guidance
6			$\frac{327}{600} \times 50$	M1(AO1.1b)	may be implied by 27.25
			27 cao	A1(AO1.1b)	
				[2]	
			Total	2	

Question			Answer/Indicative content	Marks	Guidance	
7		a	eg some people have more than one phone oe	E1(AO2.4) [1]	Some of the other figures in the LDS are greater than 1	
		b	E.g. data missing E.g. removal of outlier data items	B1(AO1.1b) B1(AO1.1b) [2]		
		c	Upper quartile	B1(AO1.2) [1]		
		d	$X \sim N(996, 407^2)$, $p = 0.75$ 1271	M1(AO3.4) A1(AO1.1b) [2]	BC allow awrt 1300	if **M0** allow **B2** for awrt 1300 unsupported
		e	*p*-value = 0.127 > 0.05 not significant Insufficient evidence to suggest that the population mean of the number of mobile phones/1000 population has increased	M1(AO1.1b) M1(AO1.1b) A1(AO2.2b) E1(AO2.2b) [4]	from screenshot comparison with 0.05 FT	
		f	Sample size of 12 is small compared to the number of countries in the population so inference may be unreliable	B1(AO2.3) [1]	Any sensible comment on the premise that 12 is a relatively small sample size	
			Total	1		

Question			Answer/Indicative content	Marks	Guidance
8		a	Any two distinct reasons eg classes of different widths represented by bars of same width eg vertical axis should be frequency density eg final upper class boundary not given eg should have continuous horizontal scale / no gaps between bars	E1(AO2.4) E1(AO1.1) [2]	**Examiner's Comments** Many candidates gave valid answers. However, the question asked for two respects in which the presentation of the data is incorrect, and an answer like 'it should be a histogram' does not answer the question.

Question			Answer/Indicative content	Marks	Guidance	
		b	(i) Positive correlation	B1(AO2.2b) [1]	oe	
					Examiner's Comments Most candidates gave a correct answer to this part. However, stating that there is a correlation between birth and death rates was not given any credit, nor were statements like 'higher birth rates cause higher death rates'.	
			(ii) 9.8395 or 9.8 or 9.84 or 9.840	B1(AO1.1) [1]		
					Examiner's Comments This question was usually answered correctly. However, a small number of candidates appeared to have read an approximate answer from the graph (usually 10) despite the clear instruction to use the equation of the line of best fit to calculate this.	
			(iii) extrapolation	B1(AO3.2b) [1]	oe	
					Examiner's Comments Many candidates correctly answered that this would require extrapolation; many others gave an acceptable equivalent answer by pointing out that there was no data in that part of the scatter diagram so using the line of best fit would be unreliable.	
			(iv) Birth rates and death rates in the Caribbean, may be very different from those in Africa.	E1(AO2.2a)	oe	Advantage

Question			Answer/Indicative content	Marks	Guidance	
				[1]		
					Examiner's Comments Most candidates were sufficiently familiar with the Large Data Set to answer this part.	
			(v) eg other continents to select countries from	E1(AO2.2a) [1]	eg a random sample would almost certainly not just include countries from Africa	Advantage
					Examiner's Comments Very few candidates answered this part correctly. Many candidates gave answers that suggested that it would be difficult to take a random sample from all the population of Africa because of communication difficulties, size of the continent, unreliability of records, etc., whilst true, do not address the context of the question. The majority of candidates did not appreciate that the sample taken was of 55 countries, and that this sample was taken from all the countries of the world, as listed in the Large Data Set.	

Question			Answer/Indicative content	Marks	Guidance
		c	Eg Generate a random number, n, between 1 and 4 and select the nth item in the data set. Eg Select every 4^{th} item on the list thereafter (stopping when 14 have been selected)	B1(AO1.2) B1(AO1.1) [2]	Candidates may choose other valid starting points Candidates may choose other valid intervals **<u>Examiner's Comments</u>** There were very few completely correct answers to this part. A great many candidates described how to find a random sample by assigning random numbers to the countries. A great many others described stratified sampling. Those candidates who did describe taking, typically, every fourth country often omitted to point out the need to use a random starting point.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
9		a	(i) the cumulative frequencies have been plotted against the mid-points of the class intervals, mis-plotting [at centre of each class] reduces estimate (by 2.5) oe	B1 (AO 2.4) B1 (AO 2.4) [2]	**Examiner's Comments** Candidates who did well in this question recognised that the cumulative frequencies had been plotted at the mid-point of the intervals instead of at the upper limit.
			(ii) grouped data has been used grouping has slightly reduced the error introduced by misplotting (because the error is less than 2.5)	B1 (AO 2.4) B1 (AO 2.4) [2]	or eg Hodge used the graph (instead of the raw data) **Examiner's Comments** Candidates understood that grouping the data affects the accuracy of the result and commented accordingly. Candidates who did less well made comments about whether the points had been joined by straight lines or a curve.
		b	percentage unemployment is often estimated oe	E1 (AO 2.4) [1]	allow data (on percentage unemployment) is not available for all countries **in Europe** oe **Examiner's Comments** Candidates who did less well based comments on general geographical or economic ideas, rather than specifically related to issues related to the estimation of median values.

Question			Answer/Indicative content	Marks	Guidance
		c	there are many other countries in the pre-release material; it is very unlikely that a random sample would only include European countries.	E1 (AO 2.4) [1]	**Examiner's Comments** Candidates who did well on part (b) and part (c) were familiar with the pre-release material made appropriate comments.
		d	negative correlation / association (may be embedded) comparison of *p*-value with 0.05 or 0.01 or other appropriate significance level and supporting comment	B1 (AO 2.2b) B1 (AO 2.2b) [2]	if **B0B0** allow **SC2** for eg comment on no significant association justified by comparison of *p* -value with appropriate significance level (eg 0.025) **Examiner's Comments** Candidates who did well commented on the nature of the correlation. They compared the p-value with a significance level and then made an appropriate deduction. Candidates who did less well compared the correlation coefficient with the p-value or did not comment on the association at all.
		e	(even though this is interpolation), the scatter / weak correlation / presence of an outlier would suggest that the use of of a line of best fit is inappropriate	E1 (AO 2.2b) [2]	allow explanation based on the value for Kosovo being an outlier or on it lying in the (large) gap in the scatter **Examiner's Comments** Candidates who did well commented on the nature of the scatter or the position of 35.3 relative to the given values to justify their comment.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
10		a	Each possible group of [pupils] (of size *n*) which could be taken from the population has the same chance of being picked	E1 (AO2.4) [1]	allow each pupil has equal chance of being sampled AND choosing one [pupil] to be sampled does not affect another being sampled
		b	eg It is not possible to select the last 23 pupils on the list. eg The different starting points on the list do not all have the same probability of being selected eg It is not possible to select some samples eg the first 50 on the list	E1 (AO2.4) E1 (AO2.4) [2]	accept any two valid reasons
		c	eg Randomly select a starting point on the list between 1 and 31. Select every 12th item on the list after that.	E1 (AO2.4) [1]	Accept any coherent statement which involves the selection of a random starting point, values being selected at regular intervals and results in all values being available for selection
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
11		a	=B2*C2	E1 (AO2.4) [3]	must have "="
		b	A All populations are in the upper quartile B All total GDPs are in the upper quartile Both statements are consistent with the data.	E1 (AO2.4) E1 (AO2.4) B1 (AO2.2a) [3]	
		c	No because some countries do not take part in the Olympics OR having more Olympic athletes in a country may encourage others	E1 (AO2.2a) [1]	
			Total	5	

Question			Answer/Indicative content	Marks	Guidance	
12		a	Opportunity sampling	B1(AO 1.2) [1]		
		b	Each sample of this size does not have an equal probability of being selected since not all the customers will come at 11.00 am on a Monday	E1(AO 2.4) [1]	allow eg Some of Qasim's customers will be at work and never be able to visit café at 11am on Monday	
		c	Positive skew	B1(AO 1.2) [1]		
		d	Median is 10^{th} value = 30 43 – 22 = 21	B1(AO 1.1) M1(AO 1.1) A1(AO 1.1) [3]	For their 15^{th} value – their 5^{th} value	NB other conventions for finding the quartiles are acceptable as long as the method is clear.
		e	Any two reasonable statements eg Medians similar, so age of typical customer found to be the same IQR much smaller in larger sample, suggesting less variability range is larger, (but these values are both outliers and could be atypical)	B1(AO 2.4) B1(AO 2.4) [2]		
			Total	8		

Question			Answer/Indicative content	Marks	Guidance
13	a		the data was not available for all countries oe	B1 (AO2.4) [1]	**Examiner's Comments** Candidates who did well in this question commented that the data may not have been available in the other three countries. Candidates who did not do well were over-imaginative in their responses, making comments on the countries being outliers or at war, but not linking this explicitly with the result that the data was not available.
	b		use of $Q_1 – 1.5×(Q_3 – Q_1)$ and $Q_3 + 1.5x(Q_3 – Q_1)$ seen for either set 4.135 < 6.28 and 15.775 > 14.46 0.38 < 3.58 and 18.86 > 14.89	M1 (AO3.1b) A1 (AO1.1) A1 (AO1.1) [3]	if **A0A0** allow **SC1** for 4.135, 15.775, 0.38 and 18.86 all seen **Examiner's Comments** Candidates who did well used the criteria for outliers based on the interquartile range and compared their results with the given smallest and largest values. Those who did less well made a slip with the arithmetic or did not make any specific comparison. Candidates who did not do well tried to work with the mean and standard deviation or were unable to use the criteria based on interquartile range, median ± 1.5 × IQR being a common error.

Question			Answer/Indicative content	Marks	Guidance
	c		22 954 isw	B1 (AO3.1b) [1]	allow 22 955, 22 950 or 23 000 NB $6411776 \times \frac{3.58}{1000}$ **Examiner's Comments** Candidates who did well were familiar with the LDS and understood that crude death rate is quoted per thousand. They gave their answer as a suitable whole number. Candidates who did not do well assumed the crude death rate was per 100 people or per day.
	d		there are almost certainly more "old" people in the population oe	B1 (AO2.4) [1]	**Examiner's Comments** Candidates who did well commented on the higher proportion of older people in Germany. Candidates who did badly commented how people might die a crude death, that Germany has a larger population or guessed at, for example, higher crime rates.

Question			Answer/Indicative content	Marks	Guidance
	e		in African countries there is a negative association / relationship between (or negative correlation between the ranks of) median age and crude death rate, but in Europe there seems to be a positive association / relationship between (or positive correlation between the ranks of) median age and crude death rate the “association” / “relationship between” or “correlation between the ranks of" median age and crude death rate (appears to be) stronger in Europe	B1 (AO2.4) B1 (AO2.4) [2]	do not allow “negative correlation” and / or “positive correlation” allow **B1** both relationships are weak oe comment comparing and contrasting type of relationship in both continents for **B1, and** one comment comparing and contrasting strength of relationship in both continents for **B1** allow equivalent explanations in words eg as median age increases crude death rates decrease in Africa and similar for Europe <u>Examiner's Comments</u> Candidates who did well in this question understood the difference between association and correlation and commented accordingly. They commented on the nature of the association and the relative strength of the association for each continent. Candidates who did not do well were in the majority. They commented on correlation and/or the strength of the linear relationship. Some candidates did not really address the question at all, instead writing commentaries on the different social and economic conditions in the two continents.
			Total	8	

Question			Answer/Indicative content	Marks	Guidance	
14	a		Quota sampling	B1 (AO1.2) [1]	**Examiner's Comments** Very few knew that the question was describing Quota sampling, though many realised it could result in bias through not being a random sample. A good many candidates knew that the answer to part (c) is 'Positive skew', though a considerable number of them described the diagram as having a peak at 1 and then tailing off. Although most candidates could find the mean in part (d), wrong answers, like 16.67, were common and few could find the standard deviation. Only a minority of candidates answered part (e) correctly. Candidates did better on part (f), partly because follow through from their answers to part (d) was allowed, although some did not put their comments in context.	
	b		non-random	B1 (AO2.4) [1]	Or equivalent	Accept 'the procedure is biased', but not 'those chosen might be biased' or 'tedious'. Ignore extra parts of explanation.
	c		Positive skew	B1 (AO1.2) [1]		
	d		mean is 1.67 sd is 1.31	B1 (AO1.1) B1 (AO1.1) [2]	CAO CAO Do not allow 1.30 or 1.3	
	e		20	B1 (AO1.1) [1]	NB $44 \times \frac{100}{220}$	

Question			Answer/Indicative content	Marks	Guidance	
	f		(means very similar so) average number of portions of fruit consumed per child is similar to average number of portions of fruits consumed per adult	B1 (AO2.2a)	Condone mean for children is (slightly) less than for adults BUT must be in context of portions of fruit	ft from (d); answers must be consistent with their answers to (d) and must both be in context
			(sd for children is lower so) less variability in number of portions of fruit consumed per day by children	B1 (AO2.5) [2]		
			Total	8		

Question			Answer/Indicative content	Marks	Guidance	
15	a		Mass = 23.56×1.816^2 =77.7 [kg]	M1 (AO1.1) A1 (AO1.1) [2]	Allow M1 for cm used here CAO **Examiner's Comments** Most candidates were able to find the mass correctly, though some did not realise the height is squared in the formula, and others used the height in cm rather than m. The correlation in part (b) was generally described correctly as 'positive' or by a description such as 'as age increases so does BMI'. Part (c) was usually done correctly, but few gave the answer that (d) would require extrapolation, which is not sensible. Some candidates did correctly explain that the data only went up to an age of 45, and so the line of best fit could not be used at 60, while others gave an incomplete argument, that the lack of data at (or about) age 60 meant the line of best fit should not be used. For part (e), a good number of candidates correctly pointed out that the sample data included no females, and the range of ages used was restricted. There were no completely correct answers to part (f); some candidates gained a mark for saying that every *n*th value should be taken (where *n* was usually 16), but none explained the need for a random starting point.	
	b		Positive [correlation]	B1 (AO1.1) [1]	Ignore, eg, weak or strong	Accept 'as age increases so does BMI' oe
	c		$28 \leq \text{BMI} < 30$	B1 (AO3.4) [1]		
	d		extrapolation	B1 (AO3.5b) [1]	Or equivalent	

Question			Answer/Indicative content	Marks	Guidance	
	e		eg females in population wider age range in population scatter diagram for whole population shows very weak negative correlation	B1 (AO2.2b) B1 (AO2.2b) [2]	any two distinct comments	
	f		Eg Generate a random number, *n*, between 1 and (eg 20) inclusive and select the *n*th item in the data set OR select every *m*th (eg 16th value). For valid solution.	M1 (AO1.2) A1 (AO1.1) [2]	Random no between 1 and 9 and every 17th item would also work.	Random no between 1 and 31, and then every 15th item also works, etc.
			Total	9		

Question			Answer/Indicative content	Marks	Guidance	
16	a		Quota sampling	B1 (AO1.2) [1]		
	b		9	B1 (AO1.1) [1]	from 5×1.8	
	c		Systematic: select every 24th number on the list start randomly between $n = 1$ and $n \geq 184$ and stop when 200 have been selected (if $n > 184$, must cycle through list) Simple random sampling: assign each item in the list a unique number (eg from 1 to 4960) generate random numbers until a sample of 200 has been selected **soi**	 M1 (AO2.4) A1 (AO1.1) E1 (AO2.4) E1 (AO1.1) [4]	*alternatively* select every 25th number on the list start randomly between $n = 1$ and $n \geq 25$, and cycle through the list again, stopping when 200 have been selected *alternatively* allow any process where each member of the population has an equal chance of being selected allow any process where each possible sample has an equal chance of being selected	*alternatively* select every 24.8th value on list, rounding as appropriate start randomly with any value on list. Cycle through the list repeatedly until 200 items have been selected
	d		as the size of the sample increases, the shape of the distribution appears more and more "Normal" **oe**	B1 (AO2.4) [1]	must refer to shape **and** closer to Normal shape for larger sample	

Question			Answer/Indicative content	Marks	Guidance	
	e		use of N(60.0515, 6.5783^2) to find P(X > 65) awrt 0.23 4960 × their 0.226 1121 or 1120 or 1119	M1 (AO3.3) A1 (AO3.4) M1 (AO3.1b) A1 (AO3.5a) [4]	condone use of 6.5717^2 or parameters rounded to 3 sf	**M0** if continuity correction used or eg P(X > 64) found
	f		eg there may be seasonal fluctuations such as teachers retiring in August	B1 (AO3.5b) [1]	allow any sensible reason in context	do not allow eg mean and sd may be different
			Total	12		
17			eg randomly select N different businesses and then randomly select P computers from each business; may be implied by correct description	B1 (AO2.4) [1]	$N \times P = 120$ where N and P are integers greater than 1.	eg 20 and 6 or 15 and 8
			Total	1		

Question			Answer/Indicative content	Marks	Guidance	
18	a		house prices are generally higher in London boroughs (than elsewhere in the country), so Dr Procter's suggestion is probably wrong	B1 (AO2.2a) [1]		
	b		214 505 219 402	B1 (AO3.4) B1 (AO1.1) [2]		
	c		$P = 28\,500Y - 57\,184\,000$ (where Y is the calendar year) or $P = 28\,500y + 215\,000$ (where y is the number of years after 2014)	B1 (AO3.3) B1 (AO1.1) [2]	gradient intercept	allow both marks for correct equation in any form **isw** allow eg $y = 28\,500x - 57\,184\,000$
	d		2016 272 000 2017 300 500	B1(AO3.4) B1(AO1.1) [2]		**FT** *their* straight line model provided this gives values > 250 000
	e		Dr Procter's model is a (very) poor fit Prof Jackson's is a good fit, or works well for 2017, but not 2016	B1 (AO2.2a) B1 (AO2.2a) [2]	dependent on correct values in (b) **FT** comment for *their* values > 250 000	this mark is dependent on having calculated values in part **(d)**
	f		neither – extrapolation **oe**	B1 (AO3.5b) [1]		
			Total	10		

Question			Answer/Indicative content	Marks	Guidance	
19	a		Lee is wrong because he should make the comparison of 0.033 with 0.05 he should make the comparison of 0.37154 with 0	 B1 (AO2.2a) B1 (AO2.2b) [2]	allow he should have compared r with the critical value	if **B0B0** **SC1** for Lee has confused r with p **or** for 0.37154 suggests positive correlation
	b		465467 + 2 × 204356 awrt 874180 (or 867940 from use of 210236) from scatter diagram the outliers are approximately 920 000, 1 200 000	M1 (AO2.1) A1 (AO2.2b) [2]	condone use of 201236 instead of 204356; ignore work relating to lower tail numerical values must be mentioned	or 521000 + 1.5 × (521000 – 342500) or 788750 in which case accept two **or** three outliers identified extra one is approximately 800 000
	c		the pmcc would (probably) be closer to 0 because the scatter is less well modelled by a straight line the p-value would increase because a value which is closer to 0 is more likely assuming there is no correlation	B1 (AO2.2b) B1 (AO2.2b) [2]	if **B0B0** allow **SC1** for r closer to 0 and p-value larger	
	d		the student's suggestion is reasonable, since there are other regions defined in the LDS	B1 (AO2.2b) [1]		
			Total	**7**		